CASE STUDY / INDUSTRIAL VIRTUALIZATION

Virtualization for SCADA engineering.

Deepak Nitrite Limited
Nandesari Division | Vadodara

VMware Workstation 17 Pro
Windows 11 Pro licensing and onsite installation
for the instrument department.

Controlled deployment. Clear ownership.

PO PRJ-20242415331 | 25 August 2023
Documented supply scope + technical implementation blueprint



Project brief | A licensed platform for instrument systems

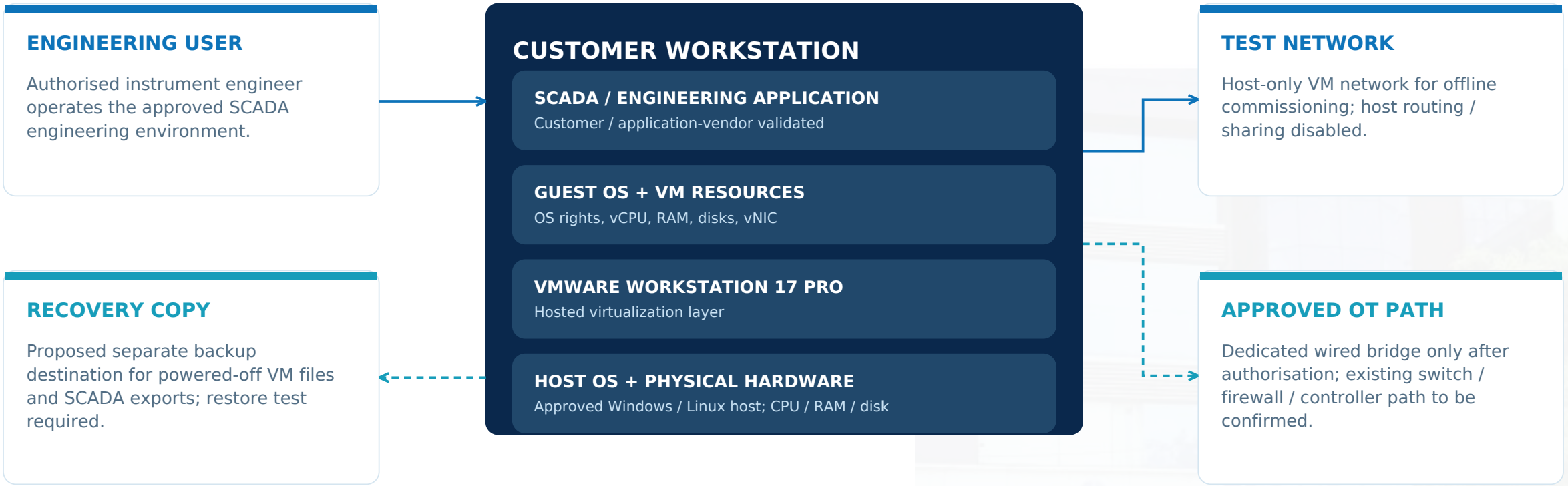
Deepak Nitrite's Nandesari Division commissioned VMware and Windows software from D-TECH for the instrument department's SCADA system. The order combines software entitlements with onsite installation support. This case study presents the documented engagement and a controlled implementation blueprint for the customer's engineering team.

Ordered scope	Quantity / reference	Delivery significance
Windows 11 Pro GGWA	5 perpetual licences; DG7GMGF0L4TL:0003	Licence allocation and genuine Windows installation / activation to be verified per device and usage rights.
VMware Workstation 17 Pro	1 licence; WS17-PRO-C	Hosted workstation virtualization, not ESXi, vCenter or a multi-host HA cluster.
Production support / subscription	1 year; WS-PRO-P-SSS-C	Historical purchase entitlement; current coverage and renewal require confirmation.
Onsite installation support	Explicitly included on PO page 2	Validate host readiness, install approved software, document configuration and obtain acceptance.

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05 / Acceptance & resilience p.06 06 / Governance & support p.07

Basis: PO dated 25 August 2023, delivery date 8 September 2023. Order value: INR 84,400 before tax; INR 99,592 including tax. No completion report supplied.

Solution architecture | The SCADA workstation stack



Illustrative guest and network paths only. Host model, VM count, SCADA version, controller interfaces, backup target and actual VLANs were not supplied.

No automatic host failover is established. Host failure can interrupt all VMs on that workstation; snapshots are not independent backups.

Micro-level deployment | Configure, record, validate

Stage	Technical implementation checks	Recorded output
01 | DISCOVERY	Inventory host CPU / RAM / storage / NIC; record OS and SCADA versions, device interfaces, dongles, dependencies and approved outage window.	Asset register; application-vendor compatibility decision; customer-approved plan.
02 | LICENSING	Map the five GGWA licences to eligible devices; verify host and guest rights separately. Install the approved Workstation build from an authorised source.	Entitlement register and activation evidence; no keys published in handover or case study.
03 | HOST READINESS	Verify virtualization support and approved BIOS settings; check disk health, capacity, patch baseline, power protection and conflicting hypervisor settings.	Host-readiness checklist; exceptions and approved remediation.
04 | VM BUILD	Set approved vCPU / RAM, disk format / size, firmware, boot mode and guest OS; install supported VMware Tools where appropriate.	VM configuration sheet; resource baseline; guest-driver verification.
05 | CONNECTIVITY	Commission on host-only networking first. Bind any approved bridge to the intended wired NIC; verify IP, mask, gateway, time source and firewall rules.	Network matrix; duplicate-IP checks; approved OT communication evidence.
06 | SCADA INTEGRATION	Install / restore vendor-approved project; verify licences, drivers, tag mappings, protocol endpoint and required USB / serial interfaces.	Vendor-led test evidence; no live control changes without authorised plant supervision.

Sizing uses measured application needs: host RAM must cover host reserve + concurrent guest allocation + headroom; storage includes growth and recovery copies.

CPU, RAM, disk and VM counts are deliberately not invented. These steps extend the installation method and require an agreed site-specific statement of work.

Legacy environment | Safe migration and controlled cutover

01 | BASELINE & EXPORT

Record existing settings; export approved SCADA projects, tag / alarm lists and relevant history.

02 | PROTECT THE SOURCE

Take an approved recoverable backup; record versions and licences. Keep the original environment unchanged.

03 | BUILD OFFLINE

Reinstall / restore into the VM. Use host-only networking; verify files, dependencies and device drivers.

04 | VALIDATE WITH OWNER

Run offline project checks and approved communication tests. Resolve USB / serial / licensing issues.

05 | APPROVED CUTOVER

Use agreed outage window; disconnect conflicting source identity before enabling any production OT path.

06 | ACCEPT OR ROLL BACK

Customer confirms results. If a gate fails, isolate the new VM and restore the approved original service path.

Migration is conditional: do not assume a legacy OS is supported or that P2V resolves drivers, dongles or application licences. Application-vendor clearance is required.

Prevent duplicate IPs and simultaneous SCADA control sessions. Backups, live commissioning and migration execution need specific customer authorisation.

Acceptance | Prove the environment before handover

Acceptance gate	Test / evidence	Owner
HOST & LICENCE BASELINE	Approved host / guest versions; assigned software entitlements; device activation; Workstation installation and VM configuration documented.	D-TECH + customer IT
APPLICATION FUNCTION	SCADA project opens; agreed displays, alarms, tags and exports match baseline. Use offline / read-only tests first.	Instrument team + application vendor
NETWORK & INTERFACES	Confirm permitted endpoints and protocols; no duplicate IP; test USB / serial / dongle behaviour where required; no unauthorised IT-to-OT route.	Instrument team + IT / OT owner
PERFORMANCE & RESTART	Measure guest memory pressure, host capacity and disk latency during representative load. Test approved reboot and operator recovery procedure.	D-TECH + customer owner
BACKUP & RESTORE	Test independent recovery copy; confirm SCADA application consistency and licence reactivation requirements. Record restore duration and limitations.	Customer backup owner + vendor
CUTOVER SIGN-OFF	Record pass / fail results, unresolved items, network activation approval and rollback decision; obtain customer acceptance.	Customer project owner

SNAPSHOT ≠ BACKUP

Use short-lived approved checkpoints where appropriate; maintain a separate recoverable copy. Do not revert live control state without plant approval.

RESILIENCE IS A SEPARATE DESIGN

No HA / DR cluster is ordered. Agree replacement-host compatibility, recovery steps and RPO / RTO before relying on this workstation for continuity.

Targets and results are not supplied; this is an acceptance checklist, not a claim of successful migration, uptime improvement or regulatory certification.

D-TECH value | Licensing, implementation and ownership

LICENCE DISCIPLINE

Connect ordered VMware and Windows software to actual usage, device allocation and documented entitlements.

CONTROLLED INSTALLATION

Bring a structured readiness check, approved configuration, offline tests and customer acceptance to onsite deployment.

CLEAR HANDOVER

Provide VM / network records, recovery dependencies, vendor responsibilities and a traceable issue register.

Support process | Register → assess → obtain approval → resolve / escalate → validate → close

Capture host asset ID, VMware build, VM name, exact symptom, time and application impact. D-TECH coordinates the agreed installation support; application issues remain with the SCADA owner / vendor. Share diagnostic evidence securely and never publish licence keys or plant credentials.

CUSTOMER / REFERENCE

Deepak Nitrite Limited, Nandesari Division
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PO PRJ-20242415331 | D-TECH named as supplier

D-TECH | CUSTOMER SUPPORT

support.dtechindia.com | Open support portal

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Source: supplied 2023 PO and client visuals. The PO establishes supply + onsite installation scope; design, migration and tests are an implementation blueprint.

Technical reference: Broadcom / VMware hosted networking types, Article 309842 | Open documentation